TESLA COACHING CLASSES

Mathematics – Part 2 (Geometry)

Time: 2 Hours	Standard: Class 10	Total Marks: 40

Part Syllabus Test - 05 (PST-05)

Student Name: _____ **Date:** _____

CHAPTERS COVERED & WEIGHTAGE

CHAPTER 3	Circle	90%
CHAPTER 1	Similarity	5%
CHAPTER 2	Pythagoras Theorem	5%

Q.1(A) **Choose the correct alternative from given:** **[4 Marks]**

i) Two circles intersect each other such that each circle passes through the centre of the other. If the distance between their centres is 12, what is the radius of each circle?

(a) 6 cm **(b)** 12 cm

(c) 24 cm **(d)** can't say

ii) In a cyclic $\square ABCD$, twice the measure of $\angle A$ is thrice the measure of $\angle C$. Find the measure of $\angle C$?

(a) 36° **(b)** 72°

(c) 90° **(d)** 108°

iii) What is the measurement of angle inscribed in a semicircle?

(a) 90° **(b)** 120°

(c) 100° **(d)** 60°

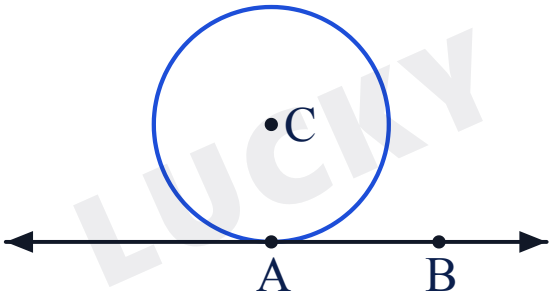
iv) How many circles can drawn passing through three non-collinear points?

(a) 0 **(b)** Infinite

(c) 2 **(d)** One and only one (unique)

Q.1(B) **Solve the following subquestions:** **[4 Marks]**

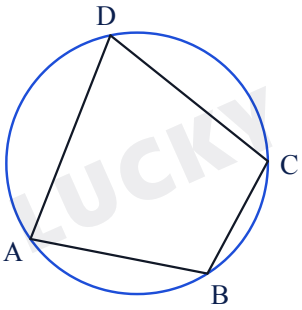
i) In the adjoining figure the radius of a circle with centre C is 6 cm, line AB is a tangent at A. What is the measure of $\angle CAB$? Why?



ii) Two circles having radii 3.5 cm and 4.8 cm touch each other internally. Find the distance between their centres.

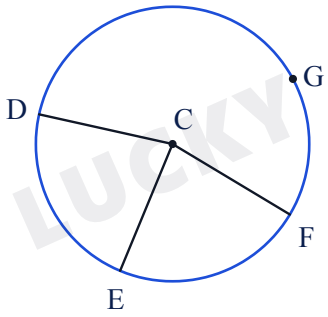
iii) What is the measure of a semicircular arc?

iv) In the figure, quadrilateral $ABCD$ is a cyclic. If $\angle DAB = 75^\circ$, then find measure of $\angle DCB$.



Q.2(A) **Complete and write any TWO activities from the following:** **[4 Marks]**

i) In figure, points G, D, E, F are concyclic points of a circle with centre C.
$\angle ECF = 70^\circ, m(\text{arc } DGF) = 200^\circ$. Find $m(\text{arc } DEF)$ by completing activity.



$m(\text{arc } EF) = \angle ECF$ ---[Definition of measure of arc]

$\therefore m(\text{arc } EF) =$ $\square$ ---(i)

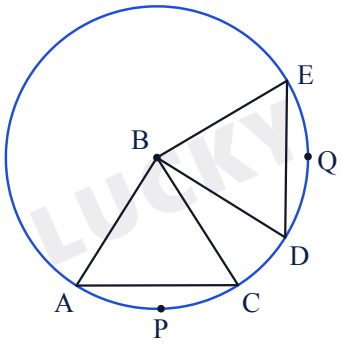
But; $m(\text{arc } DE) + m(\text{arc } EF) + m(\text{arc } DGF) =$ $\square$ ---[Measure of a complete circle]

$\therefore m(\text{arc } DE) + 70^\circ + 200^\circ = 360^\circ \implies m(\text{arc } DE) + 270^\circ = 360^\circ$

$\therefore m(\text{arc } DE) =$ $\square$ ---(ii)

$\therefore m(\text{arc } DEF) = m(\text{arc } DE) + m(\text{arc } EF) = 90^\circ + 70^\circ =$ $\square$

ii) The chords corresponding to congruent arcs of a circle are congruent. Prove the theorem by completing following activity.



Given: In a circle with centre B , arc APC ≅ arc DQE

To Prove: Chord AC ≅ chord DE

Proof: In ΔABC and ΔDBE ,

side AB ≅ side DB --- [ ]

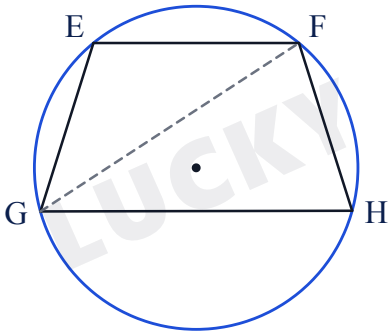
side BC ≅ side [ ] --- [ ]

∠ABC ≅ ∠DBE ---[Measure of congruent arcs]

∴ ΔABC ≅ ΔDBE --- [ ]

∴ chord AC ≅ chord DE ---[Corresponding sides of congruent triangles]

iii) In figure, chord EF ∥ chord GH. Prove that, chord EG ≅ chord FH. Fill in the blanks and write the proof.



Proof: Draw seg GF .

∠EFG = ∠FGH --- [ ] ---(I)

∠EFG = [ ] ---[Inscribed angle theorem] ---(II)

∠FGH = [ ] ---[Inscribed angle theorem] ---(III)

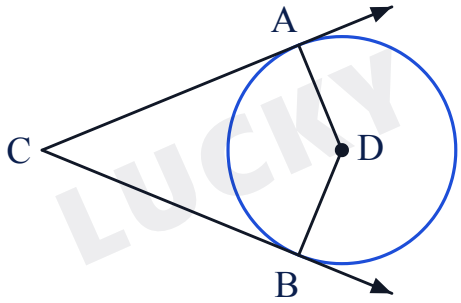
∴ m(arc EG) = [ ] ---[By (I), (II) and (III)]

chord EG ≅ chord FH ---[Corresponding chords of congruent arcs]

Q.2(B)	**Solve any FOUR subquestions from the following:**	**[8 Marks]**

i) Ratio of areas of two triangles with equal heights is 2 : 3. If base of the smaller triangle is 6 cm then what is the corresponding base of the bigger triangle?

ii) In the adjoining figure, circle with center D touches the sides of $\angle ACB$ at A and B. If $\angle ACB = 52^\circ$, find measure of $\angle ADB$.



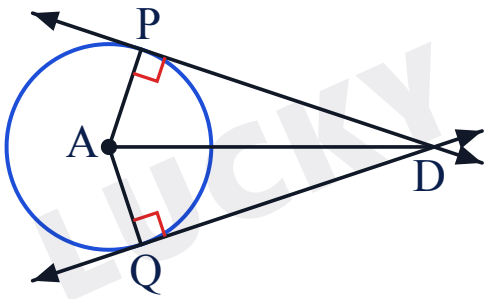
iii) In ΔPQR; $PQ = \sqrt{8}$, $QR = \sqrt{5}$, $PR = \sqrt{3}$. Is ΔPQR a right angled triangle? If yes, which angle is of 90°?

iv) Length of a tangent segment drawn from a point which is at a distance 15 cm from the centre of a circle is 12 cm, find the diameter of the circle?

v) What is the distance between two parallel tangents of a circle having radius 4.5 cm? Justify your answer.

Q.3(A)	**Complete and write any ONE activity from the following:**	**[3 Marks]**

i) Tangent segments drawn from an external point to a circle are congruent, prove this theorem. Complete the following activity.



Given: ☐

To Prove: ☐

Proof: Draw radius AP and radius AQ .

In ΔPAD and ΔQAD ,

seg $PA \cong$ seg ☐ ---[Radii of the same circle]

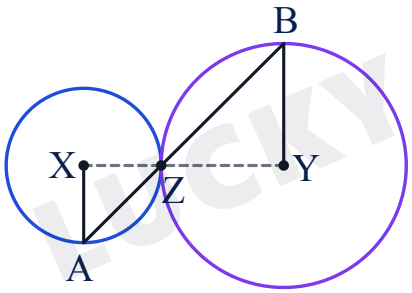
seg $AD \cong$ seg AD --- ☐

$\angle APD = \angle AQD = 90^\circ$ ---[Tangent theorem]

$\therefore \Delta PAD \cong \Delta QAD$ --- ☐

$\therefore$ seg $DP \cong$ seg DQ --- ☐

ii) In the adjoining figure circles with centres X and Y touch each other at point Z. A secant passing through Z intersects the circles at points A and B respectively. Prove that, radius XA $\parallel$ radius YB. Fill in the blanks and complete the proof.



By theorem of touching circles, points X, Z, Y are [ ] .

$\therefore \angle XZA \cong$ [ ] ---[Vertically opposite angles]

Let $\angle XZA = \angle BZY = a$ ---(i)

Now, seg XA $\cong$ seg XZ ---[Radii of the same circle]

$\therefore \angle XAZ =$ [ ] $= a$ ---[Isosceles triangle theorem] ---(ii)

Similarly, seg YB $\cong$ seg YZ ---[Radii of the same circle]

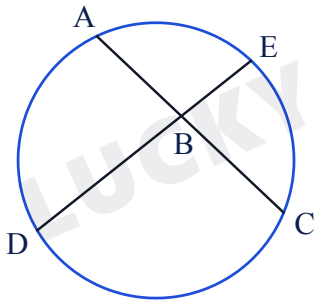
$\therefore \angle BZY =$ [ ] $= a$ ---[Isosceles triangle theorem] ---(iii)

$\therefore$ From (i), (ii), (iii), $\angle XAZ =$ [ ]

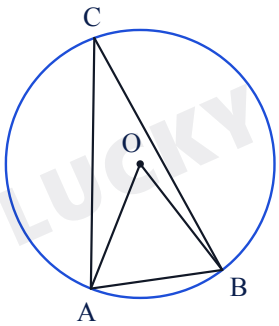
$\therefore$ radius XA $\parallel$ radius YB --- [ ]

Q.3(B)	Solve any TWO subquestions from the following:	[6 Marks]

i) In figure, chords AC and DE intersect at B. If $\angle ABE = 108°$, $m(\text{arc AE}) = 95°$, find $m(\text{arc DC})$.

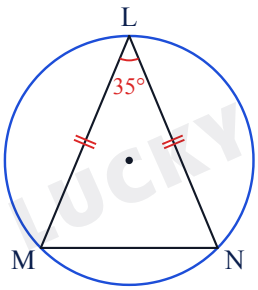


ii) In figure, in a circle with center O, length of chord AB is equal to the radius of the circle. Find measure of each of the following: (i) $\angle AOB$ (ii) $\angle ACB$ (iii) arc AB



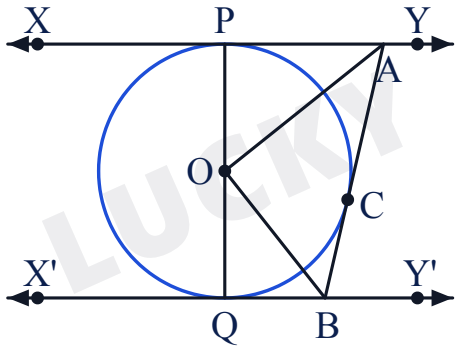
iii) **Prove the following theorem:** Angles inscribed in the same arc are congruent.

iv) In the figure, chord $\mathrm{LM} \cong$ chord LN, $\angle \mathrm{L} = 35^\circ$. Find (i) $m(\text{arc MN})$ (ii) $m(\text{arc LN})$

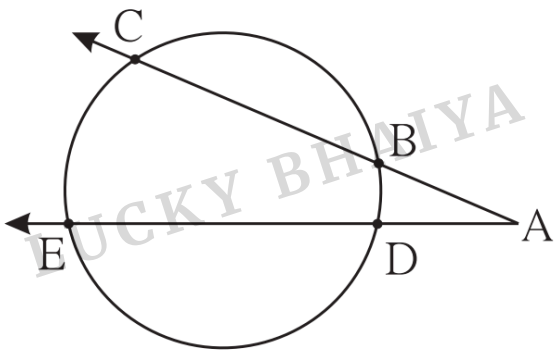


Q.4	Solve any TWO subquestions from the following:	[8 Marks]

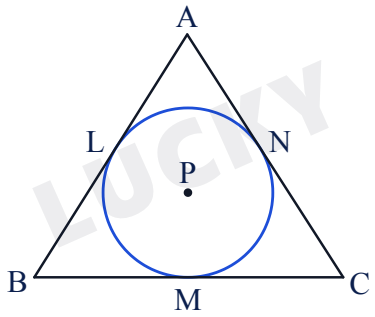
i) In the figure, segment PQ is the diameter of the circle with center O. The tangent to the circle drawn from point C on it, intersects the tangents drawn from points P and Q at points A and B respectively, prove that $\angle \mathrm{AOB} = 90^\circ$.



ii) In figure, (1) $m(\text{arc CE}) = 54^\circ$, $m(\text{arc BD}) = 23^\circ$, find measure of $\angle \mathrm{CAE}$. (2) If $\mathrm{AB} = 4.2$, $\mathrm{BC} = 5.4$, $\mathrm{AE} = 12.0$, find AD (3) If $\mathrm{AB} = 3.6$, $\mathrm{AC} = 9.0$, $\mathrm{AD} = 5.4$, find AE

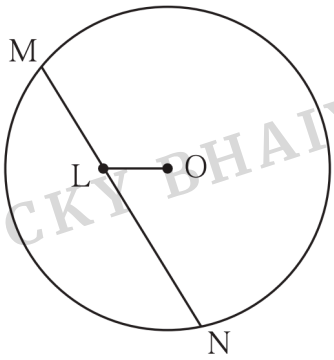


iii) A circle with centre P is inscribed in the $\Delta\mathrm{ABC}$. Side AB, side BC and side AC touches the circle at points L, M and N respectively. Radius of the circle is r. Prove that:
$A(\Delta\mathrm{ABC}) = \frac{1}{2}(\mathrm{AB} + \mathrm{BC} + \mathrm{AC}) \times r$.



Q.5	**Solve any ONE subquestion from the following:**	**[3 Marks]**

i) In figure, seg MN is a chord of a circle with centre O. $\mathrm{MN} = 25$, L is a point on chord MN such that $\mathrm{ML} = 9$ and $d(\mathrm{O}, \mathrm{L}) = 5$. Find the radius of the circle.



ii) Circles with centers A, B and C touch each other externally. If $\mathrm{AB} = 36, \mathrm{BC} = 32, \mathrm{CA} = 30$, then find the radii of each circle.

ALL THE BEST

DESIGNED BY YOUR MENTOR – LUCKY BHAIYA